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Branch	CMPN/B
NPTEL Week	4

Screenshots

```
Source
Console Terminal Background Jobs
R 4.6.1 - ~/Documents/R Programming/Assignment_4/
> x = matrix(nrow=4, ncol=3, data=c(1:12))
> x
      [,1] [,2] [,3]
[1,]    1    5    9
[2,]    2    6   10
[3,]    3    7   11
[4,]    4    8   12
>
> rownames(x) = c("r1", "r2", "r3", "r4")
> x
      [,1] [,2] [,3]
r1     1    5    9
r2     2    6   10
r3     3    7   11
r4     4    8   12
>
> colnames(x) = c("c1", "c2", "c3")
> x
      c1 c2 c3
r1     1    5    9
r2     2    6   10
r3     3    7   11
r4     4    8   12
>
> # Assign the same value to all matrix elements
> x = matrix(nrow=4, ncol=2, data=2)
> x
      [,1] [,2]
[1,]    2    2
[2,]    2    2
[3,]    2    2
[4,]    2    2
>
> # Diagonal matrices
> d = diag(1, nrow=3, ncol=3)
> d
      [,1] [,2] [,3]
[1,]    1    0    0
[2,]    0    1    0
[3,]    0    0    1
>
> d = diag(5, nrow=3, ncol=3)
```

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Console Terminal Background Jobs
R 4.6.1 - ~/Documents/R Programming/Assignment_4/
> d
      [,1] [,2] [,3]
[1,]    5    0    0
[2,]    0    5    0
[3,]    0    0    5
>
> # Transpose
> x = matrix(nrow=4, ncol=2, data=1:8, byrow=T)
> x
      [,1] [,2]
[1,]    1    2
[2,]    3    4
[3,]    5    6
[4,]    7    8
>
> xt = t(x)
> xt
      [,1] [,2] [,3] [,4]
[1,]    1    3    5    7
[2,]    2    4    6    8
>
> # Row and column sums
> x = matrix(nrow=4, ncol=2, data=c(1,2,3,4,5,6,7,8))
> x
      [,1] [,2]
[1,]    1    5
[2,]    2    6
[3,]    3    7
[4,]    4    8
>
> rowSums(x)
[1]  6  8 10 12
> colSums(x)
[1] 10 26
>
> # Row and column means
> x = matrix(nrow=4, ncol=2, data=c(1,2,3,4,5,6,7,8))
> x
      [,1] [,2]
[1,]    1    5
[2,]    2    6
[3,]    3    7
[4,]    4    8
>
```

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Console Terminal Background Jobs
R 4.6.1 - ~/Documents/R Programming/Assignment_4/

> rowMeans(x)
[1] 3 4 5 6
> colMeans(x)
[1] 2.5 6.5
>
> # Matrix indexing
> x = matrix(nrow=5, ncol=3, byrow=T, data=1:15)
>
> x[3,]
[1] 7 8 9
> x[,2]
[1] 2 5 8 11 14
> x[4:5, 2:3]
      [,1] [,2]
[1,] 11 12
[2,] 14 15
> x[c(1,4), c(1,3)]
      [,1] [,2]
[1,] 1 3
[2,] 10 12
>
> x
      [,1] [,2] [,3]
[1,] 1 2 3
[2,] 4 5 6
[3,] 7 8 9
[4,] 10 11 12
[5,] 13 14 15
>
> # Scalar operations on matrices
> x = matrix(nrow=4, ncol=2, data=1:8, byrow=T)
> x
      [,1] [,2]
[1,] 1 2
[2,] 3 4
[3,] 5 6
[4,] 7 8
> x + 5
      [,1] [,2]
[1,] 6 7
[2,] 8 9
[3,] 10 11
[4,] 12 13
>
> x = matrix(nrow=4, ncol=2, data=1:8, byrow=T)
> x

```

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Console Terminal Background Jobs
R 4.6.1 - ~/Documents/R Programming/Assignment_4/

      [,1] [,2]
[1,] 1 2
[2,] 3 4
[3,] 5 6
[4,] 7 8
> x - 5
      [,1] [,2]
[1,] -4 -3
[2,] -2 -1
[3,] 0 1
[4,] 2 3
>
> x = matrix(nrow=4, ncol=2, data=1:8, byrow=T)
> x
      [,1] [,2]
[1,] 1 2
[2,] 3 4
[3,] 5 6
[4,] 7 8
> 5 * x
      [,1] [,2]
[1,] 5 10
[2,] 15 20
[3,] 25 30
[4,] 35 40
>
> x = matrix(nrow=4, ncol=2, data=1:8, byrow=T)
> x
      [,1] [,2]
[1,] 1 2
[2,] 3 4
[3,] 5 6
[4,] 7 8
> x / 2
      [,1] [,2]
[1,] 0.5 1
[2,] 1.5 2
[3,] 2.5 3
[4,] 3.5 4
>
> # Matrix addition and subtraction
> x = matrix(nrow=4, ncol=2, data=1:8, byrow=T)
> y = matrix(nrow=4, ncol=2, data=11:18, byrow=T)
>

```

	ConsoleTerminalBackground Jobs R - R 4.6.1 - ~/Documents/R Programming/Assignment_4/ <pre> > x [,1] [,2] [1,] 1 2 [2,] 3 4 [3,] 5 6 [4,] 7 8 > y [,1] [,2] [1,] 11 12 [2,] 13 14 [3,] 15 16 [4,] 17 18 > x + y [,1] [,2] [1,] 12 14 [2,] 16 18 [3,] 20 22 [4,] 24 26 > x - y [,1] [,2] [1,] -10 -10 [2,] -10 -10 [3,] -10 -10 [4,] -10 -10 > > # Other scalar expressions > x = matrix(nrow=4, ncol=2, data=1:8, byrow=T) > x [,1] [,2] [1,] 1 2 [2,] 3 4 [3,] 5 6 [4,] 7 8 > > 4 * x [,1] [,2] [1,] 4 8 [2,] 12 16 [3,] 20 24 [4,] 28 32 > x + 4*x [,1] [,2] [1,] 5 10 [2,] 15 20 [3,] 25 30 [4,] 35 40 </pre>	
	ConsoleTerminalBackground Jobs R - R 4.6.1 - ~/Documents/R Programming/Assignment_4/ <pre> > 4*x - x [,1] [,2] [1,] 3 6 [2,] 9 12 [3,] 15 18 [4,] 21 24 > > # Matrix multiplication > x = matrix(nrow=4, ncol=2, data=1:8, byrow=T) > y = matrix(nrow=2, ncol=4, data=11:18, byrow=T) > > x [,1] [,2] [1,] 1 2 [2,] 3 4 [3,] 5 6 [4,] 7 8 > y [,1] [,2] [,3] [,4] [1,] 11 12 13 14 [2,] 15 16 17 18 > > x %*% y [,1] [,2] [,3] [,4] [1,] 41 44 47 50 [2,] 93 100 107 114 [3,] 145 156 167 178 [4,] 197 212 227 242 > y %*% x [,1] [,2] [1,] 210 260 [2,] 274 340 > > # X'X and XX' > x = matrix(nrow=4, ncol=2, data=1:8, byrow=T) > x [,1] [,2] [1,] 1 2 [2,] 3 4 [3,] 5 6 [4,] 7 8 > </pre>	

	ConsoleTerminalBackground Jobs R - R 4.6.1 - ~/Documents/R Programming/Assignment_4/ <pre> > t(x) [,1] [,2] [,3] [,4] [1,] 1 3 5 7 [2,] 2 4 6 8 > t(x) %*% x [,1] [,2] [1,] 84 100 [2,] 100 120 > x %*% t(x) [,1] [,2] [,3] [,4] [1,] 5 11 17 23 [2,] 11 25 39 53 [3,] 17 39 61 83 [4,] 23 53 83 113 > > # Cross product > x = matrix(nrow=4, ncol=2, data=1:8, byrow=T) > x [,1] [,2] [1,] 1 2 [2,] 3 4 [3,] 5 6 [4,] 7 8 > > t(x) [,1] [,2] [,3] [,4] [1,] 1 3 5 7 [2,] 2 4 6 8 > crossprod(x) [,1] [,2] [1,] 84 100 [2,] 100 120 > > # Row-wise concatenation > x = matrix(nrow=3, ncol=2, data=1:6, byrow=T) > y = matrix(nrow=3, ncol=2, data=11:16, byrow=T) > > x [,1] [,2] [1,] 1 2 [2,] 3 4 [3,] 5 6 </pre>	
	ConsoleTerminalBackground Jobs R - R 4.6.1 - ~/Documents/R Programming/Assignment_4/ <pre> > y [,1] [,2] [1,] 11 12 [2,] 13 14 [3,] 15 16 > rbind(x, y) [,1] [,2] [1,] 1 2 [2,] 3 4 [3,] 5 6 [4,] 11 12 [5,] 13 14 [6,] 15 16 > > # Column-wise concatenation > x = matrix(nrow=3, ncol=2, data=1:6, byrow=T) > y = matrix(nrow=3, ncol=2, data=11:16, byrow=T) > > x [,1] [,2] [1,] 1 2 [2,] 3 4 [3,] 5 6 > y [,1] [,2] [1,] 11 12 [2,] 13 14 [3,] 15 16 > cbind(x, y) [,1] [,2] [,3] [,4] [1,] 1 2 11 12 [2,] 3 4 13 14 [3,] 5 6 15 16 > > # Inverse > y = matrix(+ nrow = 2, + ncol = 2, + byrow = T, + data = c(84,100,100,120) +) > y [,1] [,2] [1,] 84 100 [2,] 100 120 </pre>	

ConsoleTerminalBackground Jobs

R • R 4.6.1 • ~/Documents/R Programming/Assignment_4/

> solve(y)
[1,] [1,] [2,]
[1,] 1.50 -1.25
[2,] -1.25 1.05
>
> # Eigen values and vectors
> y = matrix(
+ nrow = 2,
+ ncol = 2,
+ byrow = T,
+ data = c(84,100,100,120)
+)
> y
[1,] [1,] [2,]
[1,] 84 100
[2,] 100 120
> eigen(y)
eigen() decomposition
\$values
[1] 203.6070864 0.3929136

\$vectors [1,] [2,]
[1,] 0.6414230 -0.7671874
[2,] 0.7671874 0.6414230

>
> # --- Logical operators ---
>
> # OR: || versus |
> x = 8
> (x < 10) || (x < 2)
[1] TRUE
>
> x = 18
> (x < 10) || (x < 2)
[1] FALSE
>
> x = c(8,18)
> (x < 10) || (x < 2)
Error in (x < 10) || (x < 2) : 'length = 2' in coercion to 'logical(1)'

Show Traceback
Rerun with Debug

R Global Environment

Data

d	num [1:3, 1:3] 5 0 0 0 5 0 0 0 5	
warpbreaks	54 obs. of 3 variables	
xt	int [1:2, 1:4] 1 2 3 4 5 6 7 8	
y	num [1:2, 1:2] 84 100 100 120	

Values

x	num [1:2] 8 18
x1	num [1:4] 1 2 3 4
x2	num [1:4] 1 4 9 16